EMILY WEYDA *// Software Engineer*

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Results-driven backend engineer with 6+ years of experience designing and scaling distributed systems that power mission-critical products. Recognized for raising engineering standards, improving reliability, and fostering clarity and collaboration through strong documentation, mentorship, and a culture of technical excellence.

Professional Experience

Software Engineer, Core Services & Pipeline // Flock Safety — FEB 2024 - PRESENT

- Spearheaded the architecture and development of high-performance, scalable core services in ***Golang*** and ***TypeScript***, designing microservices capable of efficiently processing 1.3B+ real-time images daily.
- Built and launched a geospatial alert-filtering system and centralized preference service (***Go***, ***PostgreSQL***, ***DynamoDB***, ***gRPC***, ***Protobuf***), pioneering gRPC adoption and defining scalable communication patterns that became foundational to a flagship company initiative, earning recognition for deep technical ownership and product impact.
- Drove large-scale ***AWS RDS*** (***PostgreSQL***) performance optimizations, implementing deduplication and indexing strategies that cut infrastructure costs by ~90% while significantly improving latency and CPU efficiency.
- Partnered with SRE to modernize observability and CI pipelines using ***Grafana***, ***Prometheus***, ***Datadog***, ***CloudWatch***, ***OpenTelemetry*** and ***GitHub Actions***, delivering greater reliability, visibility and developer productivity.
- Established the gold standard for documentation and knowledge sharing, transforming organizational practices and earning widespread recognition for making complex systems approachable and improving engineering efficiency.
- Forged strong cross-functional relationships across Engineering, Product and Design, serving as a trusted technical liaison to ensure seamless execution and delivery of complex, high-impact initiatives spanning multiple teams.
- Championed engineering best practices and mentorship, aligning teams on scalable architecture principles and collaborative workflows, fostering a high-trust culture of technical excellence and shared ownership.

Platform Software Engineer, Core Services // Descartes Labs — FEB 2021 - SEP 2023

- Designed, developed and maintained scalable systems and subsystems powering the Descartes Labs (DL) platform, granting low-latency access to nearly 5 petabytes of geospatial data and analytics.
- Migrated critical services, including a highly scalable delayed job processing and queueing system from ***GCP*** to ***AWS***, cutting cloud costs by 70% and enabling cloud-agnostic data ingest and image processing.
- Engineered a serverless cloud computing service harnessing ***AWS*** (***S3***, ***SQS***, ***ECS***, ***CodeBuild***, ***Step Functions***) enabling users to execute arbitrary ***Python*** code efficiently at scale, 30% faster than its predecessor.
- Developed a ***Python***-based, ***gRPC***-powered parameterized model API, safeguarding intellectual property while allowing customer parameter adjustments, complete with an ergonomic client interface and seamless integration with DL's central data repository for efficient data access.
- Built a secure, ***Golang***-based structured data storage system with ***gRPC*** and ***PostgreSQL***, simplifying asset access and sharing among users and orgs across platform services, enhancing usability and user experience.
- Promoted exceptional teamwork by facilitating in the interview and onboarding processes, fostering remote collaboration and paired programming, and driving team growth in a 100% remote environment.

R&D Software Engineer // Siemens Software (Co-op) — JAN 2019 - AUG 2019

- Led the integration of ***GCP***, ***Google AutoML***, ***Docker*** and ***Node.js*** to deliver impactful proof-of-concept projects, while actively exploring machine learning, quantum computing and natural language processing technologies.

Cyber Software Engineer // ICR, Inc. (Co-op) — JAN 2018 - MAY 2018

- Researched ***ARM*** binary exploitation and reverse engineering on ***Raspberry Pi*** using ***Binary Ninja*** and ***GDB***.

R&D Software Engineer // Honeywell Intelligrated (Co-op) — AUG 2016 - AUG 2017

- Delivered customer-centric solutions within an ***Agile Scrum*** team using ***C#/.NET***, ***Microsoft SQL Server*** & ***WinForms***.

Education

University of Cincinnati // College of Engineering & Applied Science — CLASS OF 2020
Bachelor of Science in Computer Engineering, Minor in Computer Science

University of Birmingham // Study Abroad · School of Computer Science · First Class Honours — FALL 2018

Technical Skills

Languages: Golang · Python · TypeScript · SQL · Java · Bash · C · C++ · C#/.NET
Technologies: AWS · Docker · Kubernetes · Git · PostgreSQL · DynamoDB · Redis · GCP · Terraform · Bazel